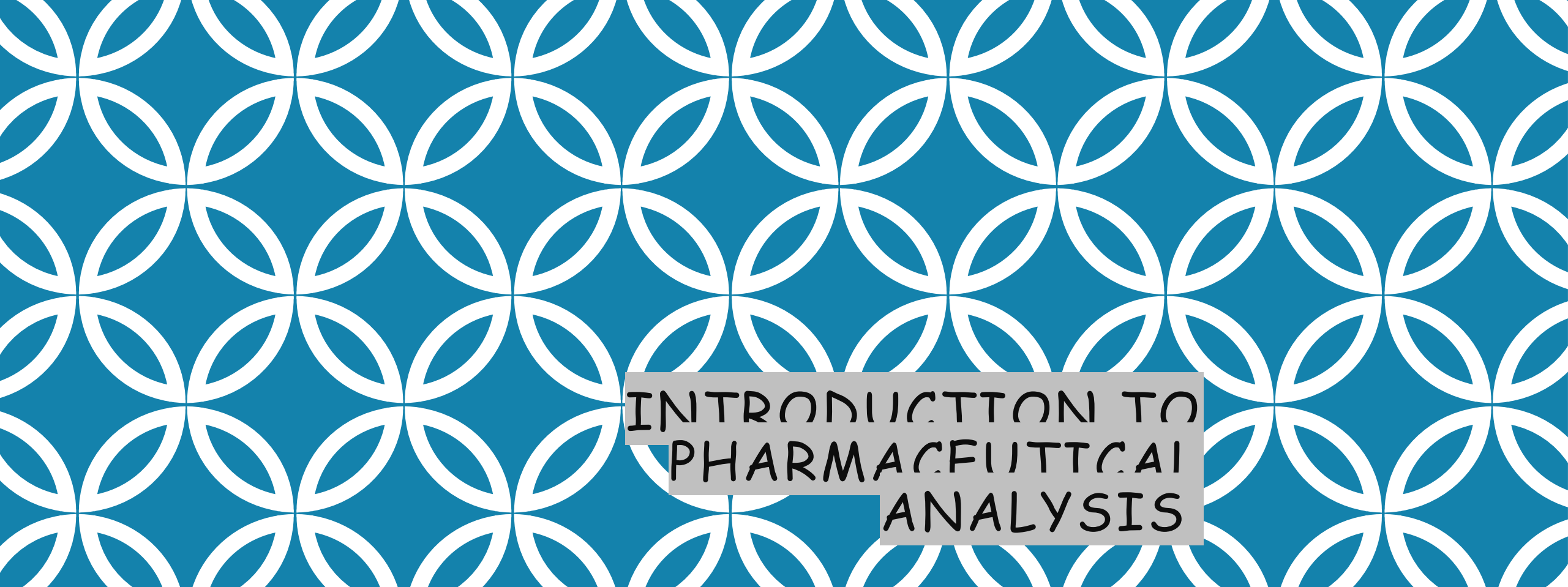


PHARMACEUTICAL ANALYSIS-I



INTRODUCTION TO PHARMACEUTICAL ANALYSIS



LEARNING OUTCOMES

After successful completion of this unit, you will be able to:

- Define what pharmaceutical analysis is and its purpose
- Scopes of pharmaceutical analysis
- Explain the central aim of pharmaceutical analysis
- Define terminologies used in pharmaceutical analysis
- Methods used for analysis
- Explain the official methods used for analysis

INTRODUCTION

- What is a Pharmaceutical?
- What is the goal of a medication?
- A pharmaceutical should be ...
- How can we ensure this?

INTRODUCTION

- A pharmaceutical
 - Is a chemical entity of therapeutic interest.
- The drug product is prepared by formulating a drug substance with inert ingredients (excipients) to prepare a drug product that is suitable for administration to patients.
- So it is the drug product that is more likely to be administered to a patient as a medicinal drug than a drug substance by itself.

An **excipient** is: any constituent of a medicinal product that is not an active substance. E.g. Adjuvants, stabilizers, antimicrobial preservatives, diluents, antioxidants

Formulation is the process in which different chemical substances, including the active ingredient and excipients are combined to produce a final medical product. It involves developing a preparation of the drug that is both stable and acceptable to the patient

The main function of excipients;

- Ensure that the preparation has a shape and size that is easy to use for the patient.
- Ensure that the active substance is optimally absorbed in the patient.
- Ensure that the preparation has an acceptable shelf life.
- Ensure that the preparation does not have an unpleasant taste or odor.
- Ensure easy production.

There is a wide spectrum of different excipients, which varies widely from preparation to preparation. Excipients of a paracetamol tablet and a paracetamol syrup

Content	Amount (mg)	Function
Tablet (weight 285 mg)		
Paracetamol	250	Active ingredient
Hydroxypropyl cellulose		Binder
Maize starch		Disintegrant
Talcum		Glidant
Magnesium stearate		Lubricant
Syrup (volume 1 ml)		
Paracetamol	24	Active ingredient
Sorbitol		Sweetener
Glycerol		Sweetener
Polyvinylpyrrolidone		Thickening agent
Saccharine sodium salt		Sweetener
Methylparabene		Preservative
Ethylparabene		Preservative
Propylparabene		Preservative
Sodium metabisulfite		Antioxidant
Citric acid		pH regulator
Sodium citrate		pH regulator
Strawberry aroma		Flavoring agent
Water		Solvent

A **Pharmaceutical** is;

- any substance or combination of substances that may be used in or administered to human beings and/or animals with a view either to **restoring, correcting or modifying** physiological functions by exerting a pharmacological, immunological or metabolic action, or to making a medical diagnosis.
- intended to furnish a pharmacological activity or other direct effect in the **diagnosis, cure, mitigation, treatment or prevention of disease.**

Quality

- fitness for purpose.
- conformity with specification.
- Is the degree to which a set of inherent characteristics of a product, system or process fulfill requirement or specifications.
- Safe, effective, high-quality, and affordable medical products are essential to positive and equitable health outcomes for all.

Factors determining quality of medicines

- Raw materials used
- Manufacturing environment
- Formulation
- Manufacturing process
- Equipment
- Technical knowhow for production and packaging of the product
- Transportation
- Storage conditions

WHERE DO WE GET SPECIFICATIONS FOR MEDICINES?

- *Quality specifications are usually set by the manufacturers and are published in **Pharmacopoeias**.*
- Medicines are specially controlled!
- Why are medicines different from other consumer products?
- Why so much regulation of the pharm. industry?

- Medicines are unlike ordinary commodities.
 - therapeutic as well as a symbolic value in most societies
- Virtually no other product is consumed by the public on such absolute trust.
 - trust that the medicine will not do them more harm than the illness.
 - This trust can only be assured if the medicine has been adequately tested.
- They are prone to **counterfeiting**.
 - *What is drug counterfeiting?*

- WHO defines counterfeiting medicine as one which is **deliberately and fraudulently mislabeled** with respect to identity and/or source.
- ✓ Generally, counterfeit Products may include:
 - Drugs with no active ingredient(s)
 - Drugs with insufficient active ingredients
 - Drugs with active ingredient(s) different from what is stated on the packages.

- Drugs without full name and address of the manufacturer
- Herbal preparations that are toxic, harmful, ineffective or mixed with the authentic medicine.
- Drugs without expiry date, or expired and re-labelled.
- Drugs not certified and registered

Global data on counterfeit drugs?

Reasons for the rapid increase of Counterfeit Medications include:

- High price of Pharmaceuticals world wide.
- Medicines are relatively easy to transport and distribute.
- Availability of technology to re-produce reasonably near identical or forgeries.
- Globalization made access to raw materials, manufacturing, supply chain easy.
- Insufficient, weak or inconsistent laws, regulations and standards.

Current and Future Risks

- Overall counterfeiting is getting worse
- Associated Problems
 - Treatment failure
 - Growth of resistance to existing anti-infectives
 - Use of illegal funds to finance further illegal manufacture of medicines
- Effects of counterfeiting:
 - Consumers
 - Health care providers
 - Drug manufacturers
 - Governments

SUBSTANDARD MEDICINES

- In 2009, the WHO defined ‘substandard’ drugs (also called ‘out of specification products’) as ‘genuine medicines produced by manufacturers authorized by the NMRA [national medicines regulatory authority] which do not meet quality specifications set for them by national standards’
- are medicines which have failed to pass the quality measurements and standards set for them.

■ **Drug must be appropriately manufactured, tested and packaged to assure that:**

- It is the correct product.
- It is the correct strength.
- It has not degraded.
- It is free from harmful impurities and micro-organisms.
- It has not been contaminated.
- It is correctly labeled.
- It is properly packaged in a suitable container.

Identifying a substandard or falsified medical product

Some falsified medical products are almost visually identical to the genuine product and very difficult to detect. However, many can be identified by:

- examining the packaging for condition, spelling mistakes or grammatical errors;
- checking the manufacture and expiry dates and ensuring any details on the outer packaging match the dates shown on the inner packaging;
- ensuring the medicine looks correct, is not discolored, degraded or has an unusual smell;
- discussing with your pharmacist, doctor or other healthcare professional as soon as possible if you suspect the product is not working properly or you have suffered an adverse reaction; and
- reporting suspicious medical products to your National Medicines Regulatory Authority.



WHAT IS PHARMACEUTICAL ANALYSIS?



It is described as the area of chemistry responsible for characterizing the composition of matter, both qualitatively (what is present) and quantitatively (how much is present).

- Is a science which deals with identification and quantification of drugs in raw materials, dosage forms and biological fluids. OR
- ensures the safety, efficacy and stability of pharmaceutical products by using physical, chemical, biological, pharmacological and biopharmaceutical methods.



In summary, a more appropriate description of pharmaceutical analysis is

“the science of inventing and applying the concepts, principles, and strategies for measuring the characteristics of chemical systems and species”.

❖ *Main goal of pharm. Analysis* – to ensure the delivery of **quality drug product to patients** and therapeutic drug monitoring *via* analysis of pharmaceutical **raw materials, intermediate products, finished products, & biological fluids** via strict **control of production system.**

SCOPE OF PHARMACEUTICAL ANALYSIS

Drug Analysis and assay play important roles in the:

Quality Control

- Identification
- Quantification
- Purity Assessment
- Formulation analysis
- Regulatory compliance
- Development (Drug Research)
- Manufacture and
- Therapeutic use of drugs

SCOPE OF PHARMACEUTICAL ANALYSIS

The Method of drug analysis(for official ones) are described in Pharmacopeias

Modern Pharmaceutical Analysis Entails the following activities:

- Discovery of NCE and high-throughput Screening
- Production and process control
- Packaging and labeling controls
- Laboratory Control
- Degradation and impurity analysis

SCOPE OF PHARMACEUTICAL ANALYSIS

- Pre-formulation analysis
- Development of new dosage forms
- Compendial testing
- Method Development
- Setting Specifications
- Method Validation
- Stability studies
- Documentation and inspection

Pharmaceutical Analyst Scope:

- Pharmaceutical industries (R&D, Intermediate testing, raw material and product control)
- Regulatory agencies (EFDA, FDA, ...)
- Therapeutic drug monitoring (TDM)
- Research (drug development, post marketing surveillance)
- Forensics
- Doping test

Analytical Procedures may be used to answer any of the following questions:

- Is the identity of the drug in the formulated product correct?
- What is the % of the stated content of a drug present in a formulation?
- Does the sample contain solely active ingredient or additional impurities present?
- What is the stability of a drug and hence the shelf-life of the product?

- At what rate is the drug released from its formulation?
- Do the identity and purity of a drug substance meet specification?
- Do the identity and purity of excipients meet specifications
- What are the concentrations of specified impurities in the pure drug substance?
- What is the concentration of the drug in a sample of tissue or biological fluid?
- What are the pka value(s), partition coefficients, solubility, and stability of a drug substance under development?

Classification of Methods of Analysis

- Methods of Analysis, can be classified in to three based on the view of *pharmaceutical analysis*.
 - Classical methods of analysis
 - Instrumental methods of analysis
 - Biological methods of analysis

1. Classical methods of analysis

a. Volumetric methods

- assay is based on the **measurement of volume of solution** of known strength that is required to react completely with the substance to be analyzed.
- on the basis of this measurement of **volume, quantity of pure component is estimated.**
- preferred to gravimetric processes especially because of **speed and convenience.**

- This method is classified in to different types **depending up on the type of reactions** involved in the titration process.
- They are:
 - Neutralization titrations
 - Non-aqueous titrations
 - Precipitation titrations
 - Oxidation reduction titrations
 - Complexometric titrations

b. Gravimetric methods

- is a quantitative analysis **by weight**
- it is **a process of isolating and weighting** the compound of known composition i.e. purest form.
- The separation of compound is achieved by number of ways like:
 - precipitation
 - Volatilization

C. Gasometric methods:

- involve measurement of the volume of gases.
- They measure the:
 - ✓ volume of gas liberated in the given chemical reaction under the conditions that are described in the process. Or
 - ✓ Decrease in the volume of gas when a suitable agent is placed to absorb one of the gases present

2. Instrumental Methods

- are based on the relation between the content and corresponding physicochemical and physical properties of the chemical system being analyzed.

A. Methods based on appropriate electrical measurements (Electrochemical methods)

- Deals with the interaction between electrical energy and chemical change
- Examples
 - Voltammetry
 - Amperometry
 - Conductimetry

B. Methods based on Measurement of certain optical properties ;(optical methods)

- Light absorption methods :
 - ✓ UV-visible spectrophotometric methods
 - ✓ Infrared spectrometry
 - ✓ Atomic absorption spectroscopy.
- Light scattering methods:
 - Turbidimetric and nephelometric methods
- Light emission methods:
 - ✓ Emission spectroscopy
 - ✓ Flame photometry
 - ✓ Fluorimetry

C. Methods based on physical separation (chromatographic methods)

- Paper Chromatography
- Gas chromatography
- Thin Layer Chromatography
- High Pressure Liquid Chromatography

Other methods:

- Radioactivity methods
- X-ray methods
- Mass spectrometry
- light deviation methods (polarometry and refractometry)
- Thermal analysis

3. Biological methods

Commonly used for,

- Microbial Limit Tests
- Biological assay of pharmaceuticals and other materials
- Microbial Quality Control
- Preservatives and Preservative efficacy testing
- Sterility Tests

❖ All the above methodologies can be classified into qualitative and quantitative analysis based on the information needed

QUALITATIVE ANALYSIS

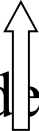
- Reveals **the identity** of the elements & compounds in a sample
- Provides presence/ absence information only

QUANTITATIVE ANALYSIS

- Indicates **the amount** of the analyte in a sample.
- Deals with how much of one or more constituents are present in a sample.
- Expressing the amount of drugs (A.I) in pharmaceutical preparation and biological fluids in numerical terms
 - **%Purity = $\frac{\text{Practical value}}{\text{Theoretical}} \times 100$**
 - Practical – measured amount
 - Theoretical – labeled amount
- It determine the percentage of purity of the drug substance or content of the active ingredients in the formulation.
- Provides a measure of concentration (a number plus units).

Remark, Qualitative analysis (information) is required before a Quantitative analysis can be undertaken.

Example,

Cimetidine	200 mg	tab.
 Identity	 Amount	 Dosage form

Therefore, qualitative methods of drug analysis is applied to identify whether the given sample of tablet is Cimetidine or not and

Quantitative methods of analysis are employed to determine the exact amount of active ingredient (Cimetidine) incorporated in a tablet

In general the two major questions in pharmaceutical analysis are,

What drug is present in a given unknown sample (qualitative analysis)

How much of the drug is present in the sample (quantitative analysis)

Drug analysis – means identification, characterization and determination of drugs

Drug assay – refers to determination of drugs in mixtures such as dosage forms and biological fluids.

Bulk drug – drug used in formulation such as syrups, tablets, and aerosol preparations.

Batch (or lot): Quantity of starting material, packaging material, or product processed in a single process or series of processes so that it is expected to be homogeneous.

Batch number (or lot number): A distinctive combination of numbers and/or letters which uniquely identifies a batch on the labels, its batch records and corresponding certificates of analysis, etc.

TERMINOLOGIES...

Batch records

- All documents associated with the manufacture of a batch of bulk product or finished product.
- They provide a history of each batch of product and of all circumstances pertinent to the quality of the final product

Finished product: A finished dosage form that has undergone all stages of manufacture, including packaging in its final container and labelling.

In-process control: Checks performed during production in order to monitor and, if necessary, to adjust the process to ensure that the product conforms to its specifications.

- The control of the environment or equipment may also be regarded as a part of in process control.

TERMINOLOGIES...

Specification

- Lists of detailed requirements with which the products or materials used or obtained during manufacture have to conform.
- They serve as a basis for quality evaluation.

Validation

- **Action of proving**, in accordance with the principles of GMP, that any procedure, process, equipment, material, activity or system actually leads to the expected results.

TERMINOLOGIES...

Standard operating procedure (SOP)

- An authorized written procedure giving **instructions for performing operations** not necessarily specific to a given product or material (e.g. equipment operation, maintenance and cleaning; validation; cleaning of premises and environmental control; sampling and inspection).

TERMINOLOGIES AND DEFINITIONS

Quality Control- involves the process of ensuring that pharmaceutical products are consistently produced and controlled to meet quality standards. It includes:-

1. Testing Raw-Materials:
2. In-Process testing
3. Final Product Testing
4. Stability Testing

QUALITY CONTROL...

- Regulatory process which measures the quality performance of products that may tend to deficiencies.
- It is carried out to monitor **deficiencies** or **defects** and achieve the goal of planning.
- The activities of QC involve laboratory procedures (i.e. physicochemical tests on raw materials, dosage forms and packing components; pharmacological tests, test for potency, sterility, microbiological status of the product, etc).

QUALITY CONTROL...

- Objectives of a QC department:
 - Make sure that proper sampling and analytical test are done
 - Make sure that the finished products contain API complying with the qualitative and quantitative composition, are of the required quantity, and enclosed within their proper container and correctly labeled.
 - Make sure that no batch of product is released to sale or supply prior to certification by qualified person.
 - Make sure that sufficient reference sample of starting materials and products are retained to permit future examination of the product if necessary.

QUALITY CONTROL

Advantages of controlling the product quality:

- Quality of product is improved which in turn increases sales
- Scrap rejection and rework are minimized thus reducing wastage. So that cost of manufacturing reduces.
- Good quality product improves reputation (status)
- Uniformity in quality can be achieved
- Improvement in technical knowledge and engineering data for process development and manufacturing design

Quality assurance

Is a responsibility of an organization to determine that *systems, facilities and written procedures both are adequate and followed in order to assure that* products are controlled and will meet in the final dosage form, all the applicable specification.

QUALITY ASSURANCE...

- ❑ It is a wide ranging concept that covers all matters that individually or collectively influence the quality of the product.
- ❑ **It is the totality of arrangements made with the objective of ensuring that pharmaceutical products are of the quality required for their intended use.**
- ❑ It incorporates GMP and other factors including product design and development.
- ❑ Therefore, QA is not the duty of one organizational unit in the company alone but is the responsibility of all staff members who in any way can influence product quality.

QUALITY ASSURANCE...

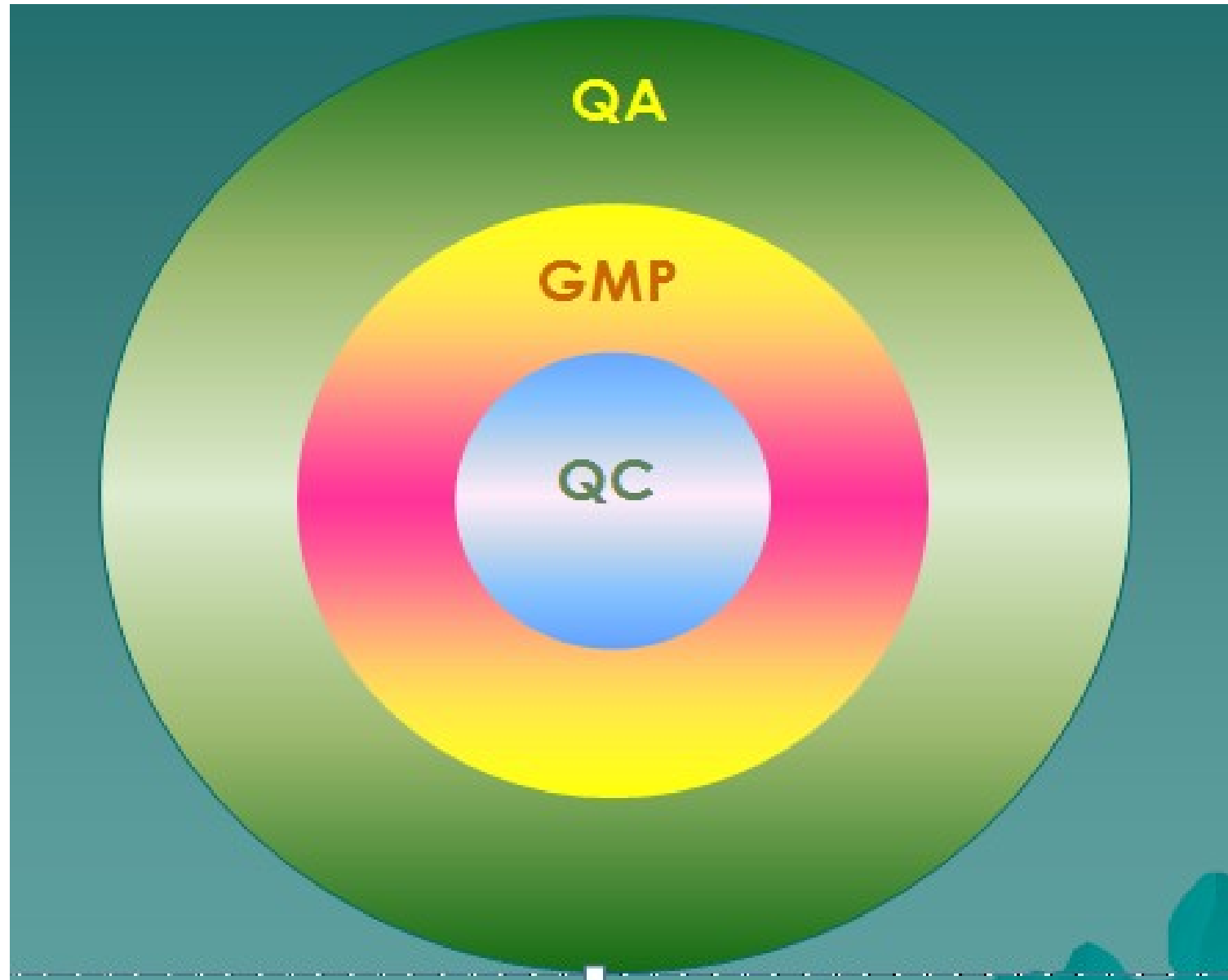
Responsibilities of quality assurance department

- QA unit is responsible for monitoring that the facilities, equipment, personnel, methods, practices, records, and controls are in compliance with the regulations.
- Plan, develop and establish quality policies.
- To assure that products of prescribed specification reaches to customers
- Regularly evaluate the effectiveness of the quality programs.
- Liaise with different departments, in and outside the organization.
- Organize training programs on quality.
- Develop quality assurance system and regularly evaluate its effectiveness.

GMP is that part of QA which ensures products are **consistently produced and controlled** to the quality standards appropriate to their intended use and as required by the marketing authorization or product specification.

GMP is concerned with both production and QA.

Is the minimum requirement that the industry must meet, when processing, manufacturing, packing, holding drugs.



GOOD MANUFACTURING PRACTICE

Requirements for GMP

- ❖ There should be clearly defined and systematically reviewed manufacturing processes.
 - All batch documents, QC specifications and relevant SOPs must be prepared in harmony with each other.
 - All departments involved must be aware of the work of other departments

GOOD MANUFACTURING PRACTICE

- ❖ Critical steps of production must be validated
 - Checking and documenting variability (materials, process, equipment)
 - Increases knowledge about the company and controlled experiment can be performed and produce products fulfilling requirements.
 - Validation is required if the process, material or equipment is new or there is any change in any part.

GOOD MANUFACTURING PRACTICE

- ❖ All necessary facilities are provided.
 - Appropriately trained and qualified personnel.
 - Adequate premises and space
 - Suitable equipment and services
 - Correct materials, containers, labels
 - Approved procedures and instructions
 - Suitable storage and transport
 - Adequate personnel, equipment, laboratories for in-process

GOOD MANUFACTURING PRACTICE

- ❖ Manufacturing is based on clearly written procedures
 - Instructions and procedures are written in unambiguous language.
 - Operators are trained
 - Complete records of the manufacturing process
 - Failure investigations are carried out if quality problems come up.

GOOD MANUFACTURING PRACTICE

- Proper storage and distribution of the products
- A recall system providing a final safety net in case quality problems are detected after release of the product
- Compliant handling procedures are established to react to feedback from the market

GOOD LABORATORY PRACTICES (GLP)

- One of the components of QA, that is concerned with the activity of the quality control laboratory carrying out test and assay required to establish the compliance of pharmaceutical or other products with the specification claimed for them.
- The quality of products is achieved through GMP and monitored if there is GLP guideline.

GOOD LABORATORY PRACTICES (GLP)

- The quality and relevance of analysis result depend on GLP guideline requiring the following test.
 - Instrument test and calibration
 - Complete validation of analytical result
 - Verification and checking analytical result
 - Control chart for trend recognition(log book for instrument., methods SOP, test data note books etc).

- **Monograph:** A written description of the principal features of the substance and the ways that these features can be determined.
- Pharmacopoeia:
 - Monographs of pharmaceutical substances when collected together.
 - Books of standards for pharmaceuticals and medical devices containing specifications and procedures of tests are known as **pharmacopoeias** and **formularies**.
 - They are collectively refereed as the drug **compendia**

PHARMACOPOEIA

- ***Pharmacon*** meaning ‘ drug’ and ‘*poiein*’ meaning ‘ make’ and the combination means any instructions of formula or other standard required to make or prepare a drug.

It is a book containing a list of medicinal substances(drugs), their dosage forms or devices with

- Descriptions of test methods,
- Specifications for purity and strength accepted by some recognized authority.

PHARMACOPOEIA

- In most countries the books of standards, is issued by the government.
 - But U.S. Pharmacopeial, private, NGO In U.S.A
- Examples:
 - United states phrmacopeia (USP)
 - British pharmacopeia (BP)
 - International pharmacopoeia (IP)
 - European pharmacopoeia(EP)
 - Japan pharmacopoeia (JP), etc.

PHARMACOPOEIA

- Reasons for having a standard (Pharmacopoeia):
 - To protect the user (patient, pharmacist, manufacturer, government office, health authority) against drugs of poor quality.
 - To guarantee /ensure that the analytical methods used are standard, keep abreast of recent developments both in analytical techniques and in quality considerations of the drug in question.

PHARMACOPOEIA...

1. The United States Pharmacopeia (USP)

- Pharmacopoeia promotes the public health and benefits practitioners and patients by disseminating authoritative standards and information developed by its volunteers for medicines, other health care technologies, and related practices used to maintain and improve health and promote optimal health care delivery Working with many constituencies and stakeholders around the world.
- USP's standards are also recognized and used in more than 130 countries.

PHARMACOPOEIA...

2. British pharmacopoeia(BP)

- The BP is the authoritative collection of standards for UK medicinal substances produced under the auspices of the medicines and healthcare products regulatory agency
- it is an essential reference point for everyone involved in research, development and manufacture of formulated (medicinal) preparations and veterinary substances.
- It is often adopted by countries all around the world that do not publish their own standards.

PHARMACOPOEIA...

- **Official monographs of pharmaceutical chemicals generally embrace the following:**
 - Description (and title)
 - Minimum standard of purity
 - Identification tests
 - Limit tests to exclude excessive impurities
 - Physical constants
 - Storage conditions and packaging
 - Labeling
 - Dosage
 - Therapeutic category
 - Quantitative assays

Titles:

Main titles

Subsidiary titles or synonyms

- Molecular formula and
- molecular weights
- Chemical names have also been provided (IUPAC)

Identification:

Tests for identity are provided only as an aid to identification.

Identifications tests are usually based on qualitative tests for **basic and acidic radicals for inorganic chemicals**.

Organic substances are identified by the various characteristic reactions of one or more of the **functional groups present in the molecule**

Dosage:

Doses mentioned in monographs are intended for general guidance

Minimum standards of purity, Assay Tolerance:

The limits are expressed numerically; the upper and lower limits of a range are inclusive so that the range consists of two values themselves and all intermediate values but no values outside the limits.

- Allow for;
 - **analytical errors**
 - **variations in manufacture and compounding, and**
 - **for deterioration to an extent considered insignificant under practical conditions**

Description:

Is relatively general in nature

It is provided to indicate the properties of the chemicals.

The properties are not in themselves standards or tests for purity
though they may help in the preliminary evaluation of chemical.

Packaging, storage and labeling:

The container should

- not interact physically or chemically with the chemical
- well-closed, etc...

Storage conditions

- *Cold*: any T° not exceeding 8° and usually between 2° and 8°C .
- *Cool*: Any T° between 8° and 25°C .
- *Room Temperature*: T° prevailing in working area
- *Warm*: Any T° between 20° and 40°C .
- *Excessive heat*: Any T° above 40°C .

Where no specific conditions are indicated it is to be understood that storage conditions includes **protection from moisture, freezing and excessive heat**.

In general labeling of drugs and pharmaceuticals is governed by **Dugs and Cosmetic Act**.

Additional information which must be stated on label is mentioned in the monograph.

Quantitative Assays:

Assay methods

- specific for the chemical

the procedures of quantitative analytical chemistry are applied to the analysis of materials used in pharmaceuticals.

Ampicillin monograph as described in BP

$C_{16}H_{18}N_3NaO_4S$ 371.4 69-52-3

Action and use

Antibacterial.

Preparation

Ampicillin Injection

DEFINITION

Ampicillin sodium contains not less than **91.0** per cent and not more than the equivalent of **100.5** per cent of sodium ampicillin

CHARACTERS

A white or almost white powder, hygroscopic, freely soluble in water, sparingly soluble in acetone, practically insoluble in fatty oils and in liquid paraffin.

IDENTIFICATION

Using TLC (described in details in the monograph)

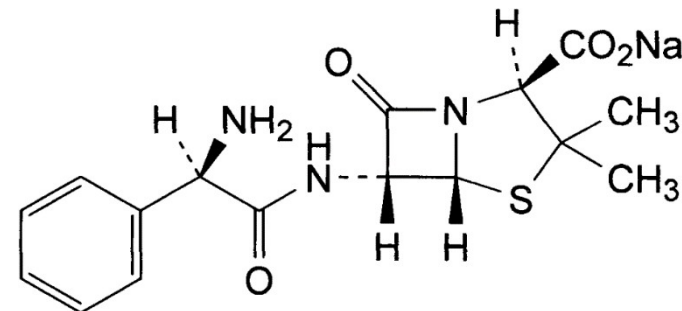
TESTS

Appearance of solution

Specific optical rotation

ASSAY

Examine by liquid chromatography





STORAGE

Store in an airtight container . If the substance is sterile, store in a sterile, airtight, tamper-proof container .

LABELLING

The label states, where applicable, that the substance is free from bacterial endotoxins.

IMPURITIES

Describes all potential impurities and their accepted limit generated from synthesis, intermediates, raw material or chemical degradation of ampicillin

Example: heavy metals not more than 20 ppm

ON THE BASIS OF NATURE OF INFORMATION: ANALYTICAL METHODS:

Proximate analysis (determine amounts of certain elements approximately).

Partial analysis (determination of selected constituents).

Trace analysis (specified constituent present in very minute quantity).

Complete analysis (determine the exact proportion of each component in the sample).

ON THE BASIS OF SAMPLE SIZE: ANALYTICAL METHODS

Macro analysis (0.1 g or more)

Semimicro analysis (0.01 - 0.1 g)

Micro analysis (0.001 - 0.01 g)

Submicro analysis (1×10^{-4} – 1×10^{-3} g)

Ultramicro analysis (Less than 1×10^{-4} g)

SAMPLE CONSTITUENTS : ANALYTICAL METHODS

Major constituent – is one present in 1-100% of sample under investigation.

Minor constituent – is one present in 0.01 - 1% of the sample.

Trace constituent – is one present in concentration $< 0.01\%$.

SOURCES OF ANALYTICAL METHODS:

- in-house methods developed by one laboratory for their own special needs.
- Method published in open scientific literature
- Methods supplied by trade organizations
- Methods in books published by professional organizations
- Methods from international organizations (e.g. ISO)
- Methods from statutory publications

FACTORS CONSIDERED IN CHOOSING ANALYTICAL METHOD

- Concentration range of ingredients
- Sensitivity of the method and the limits of detection and quantification
- The required degree of accuracy and precision
- Speed of analysis and its convenience with its purpose
- Equipment required and their availability
- Sample size and number of samples to be analyzed
- Cost
- Safety measures